

APPLIED MATHEMATICS FOR ENGINEERS - I (25MAT11)

Module -1: Elements of Linear Algebra

Elementary transformations, Reduction of the given matrix to echelon form, Rank of a Matrix, Consistency of a system of linear equations and solution. Solution of a system of linear homogeneous equations (trivial and non-trivial solutions), Solution of a system of non-homogeneous equations by Gauss-Seidel method.

Applications	Consistency in word problems.
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PART-A

- Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$ by reducing into echelon form.
- Find the rank of the matrix $\begin{bmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 5 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix}$ by reducing into echelon form.
- Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 6 & 8 \\ 4 & 8 & 12 & 16 \\ 1 & 2 & 3 & 4 \end{bmatrix}$ by using row elementary transformation.
- Find the rank of the matrix $\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$
- Find the rank of the matrix $\begin{bmatrix} 4 & 0 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 2 & 3 & 4 & 7 \\ 2 & 3 & 1 & 4 \end{bmatrix}$
- Find the rank of the matrix $\begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$
- Test for consistency and solve: $x + y + z = 6$, $x - y + 2z = 5$, $3x + y + z = 8$.
- Show that the equations $x + 2y + 3z = 14$, $4x + 5y + 7z = 35$, $3x + 3y + 4z = 21$ are consistent and hence solve them.
- Test for consistency & solve: $x - 4y + 7z = 14$, $3x + 8y - 2z = 13$, $7x - 8y + 26z = 5$
- Test for consistency & solve: $5x + 3y + 7z = 5$, $3x + 26y + 2z = 9$, $7x + 2y + 10z = 5$.
- Test for consistency and solve, $4x - 5y + z = -3$, $2x + 3y - z = 9$, $3x - y + 2z = 5$, $x + 2y - 5z = -9$.
- Test for consistency and solve, $2x + y - z = 0$, $x + y + z = 9$, $2x + 5y + 7z = 52$
- Investigate for what values of λ , μ the simultaneous equations $x + y + z = 6$, $x + 2y + 3z = 10$ & $x + 2y + \lambda z = \mu$ have (i) Unique solution (ii) Infinite solution (iii) No solution

14. Investigate for what values of λ, μ the equations $2x + 3y + 5z = 9, 7x + 3y - 2z = 8$ & $2x + 3y + \lambda z = \mu$ have (i) Unique solution (ii) Infinite solution (iii) No solution
15. Solve the system of equations $x + y + 3z = 0, x + 5y + z = 0, 3x + y + z = 0$
16. Solve the equations $x + y + 3z = 0, 2x + 3y + z = 0, 4x + 5y + 4z = 0, x + y - 2z = 0$
17. Solve the system of equations
 $x_1 + 2x_3 - 2x_4 = 0, 2x_1 - x_2 - x_4 = 0, x_1 + 2x_3 - x_4 = 0, 4x_1 - x_2 + 3x_3 - x_4 = 0.$
18. Solve the equations: $x + y + w = 0, y + z = 0, x + y + z + w = 0, x + y + 2z = 0.$
19. Solve the system of equations $x + y - 3z + 2w = 0, 2x - y + 2z - 3w = 0,$
 $3x - 2y + z - w = 0, -4x + y - 3z + w = 0$
20. If the system of equations $x + y + 3z = 0, 4x + 3y + \lambda z = 0, 2x + y + 2z = 0$ have a nontrivial solution, then find the value of λ ?
21. Solve the following system of equations
 $x_1 + x_2 - x_3 + x_4 = 0, x_1 - x_2 + 2x_3 - x_4 = 0, 3x_1 + x_2 + x_4 = 0$
22. Find the value of k such that the following system of equations possesses a non-trivial solution. Also find the solution of the system.
 $4x_1 + 9x_2 + x_3 = 0, kx_1 + 3x_2 + kx_3 = 0, x_1 + 4x_2 + 2x_3 = 0$
23. Find all the solutions of the following system of homogeneous equations
 $x_1 - 2x_2 + x_3 - x_4 = 0, x_1 + x_2 - 2x_3 + 3x_4 = 0, 4x_1 + x_2 - 5x_3 + 8x_4 = 0,$
 $5x_1 - 7x_2 + 2x_3 - x_4 = 0$
24. If the system of equations $x + y + 3z = 0, 4x + 3y + \lambda z = 0, 2x + y + 2z = 0$ have a nontrivial solution, then find the value of λ ?
25. Solve the system of equations by Gauss-Seidel method
 $10x + y + z = 12, x + 10y + z = 12, x + y + 10z = 12$
26. Solve by Gauss-Seidel method
 $x + y + 54z = 110, 27x + 6y - z = 85, 6x + 15y + 2z = 72$
27. Solve by Gauss-Seidel method: $5x + 2y + z = 12, x + 4y + 2z = 15, x + 2y + 5z = 20$
28. Solve by Gauss-Seidel method:
 $20x + y - 2z = 17, 3x + 20y - z = -18, 2x - 3y + 20z = 25.$
29. Solve by Gauss-Seidel method
 $28x + 4y - z = 32, 2x + 17y + 4z = 35, x + 3y + 10z = 24$
30. Solve by Gauss-Seidel method
 $10x - 2y + z = 12, x + 9y - z = 10, 2x - y + 11z = 20$

PART-B

1. Determine the rank of the following matrix:
$$\begin{bmatrix} 91 & 92 & 93 & 94 & 95 \\ 92 & 93 & 94 & 95 & 96 \\ 93 & 94 & 95 & 96 & 97 \\ 94 & 95 & 96 & 97 & 98 \\ 95 & 96 & 97 & 98 & 99 \end{bmatrix}$$
2. Find the rank of the matrix
$$\begin{bmatrix} 0 & 2 & -2 & 3 \\ 2 & 5 & -4 & 6 \\ -1 & -3 & 2 & -2 \\ 2 & 4 & -1 & 6 \end{bmatrix}$$
3. Test for consistency and solve, $x - 2y + 3z = 2$, $3x - y + 4z = 4$, $2x + y - 2z = 5$
4. Test for consistency and solve, $x + y + z = 9$, $x - 2y + 3z = 18$, $2x + y - z = 3$
5. Solve the system of equations $x_1 - x_2 - x_3 - x_4 = 0$, $x_1 - x_2 + x_3 + x_4 = 0$, $4x_1 - 4x_2 + 4x_3 = 0$, $-2x_1 + 2x_2 - 2x_3 + x_4 = 0$
6. If the system of equations $x + 2y - 3 = 0$, $2x - y + 4 = 0$, $\lambda x + y - 3 = 0$ are consistent then find the value of λ ?
7. Test for consistency and solve the equations $4x + 2y + z + 3w = 0$, $6x + 3y + 4z + 7w = 0$, $2x + y + w = 0$
8. Test for consistency and solve the equations $x - 2y + 3t = 2$, $2x + y + z + t = -4$, $4x - 3y + z + 7t = 8$
9. Solve the system of equations by Gauss-Seidel method $x + 17y - 2z = 48$, $2x + 2y + 18z = 30$, $30x - 2y + 3z = 48$
10. Solve the system of equations by Gauss-Seidel method $9x - y + 2z = 9$, $x + 10y - 2z = 15$, $2x - 2y - 13z = -17$

PART-C

1. Find k if rank of matrix $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & k \\ 1 & 4 & 10 & k^2 \end{bmatrix}$ is 2.
2. Find the rank of the matrix $\begin{bmatrix} 8 & 2 & 1 & 6 \\ 2 & 1 & 0 & 1 \\ 3 & 0 & 1 & 3 \\ 5 & 1 & 1 & 4 \end{bmatrix}$
3. Find the value of k such that the matrix $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & k \\ 1 & 4 & 10 & k^2 \end{bmatrix}$ is 3.
4. Solve the system of equations
 $x + 2y + 3z + w = 10, 2x + 3y + 4z + 5w = 20, 3x + 4y + 5z + 6w = 30,$
 $4x + 5y + 6z + 7w = 40$
5. Test for consistency and solve, $x + y + z = 9, 2x - 3y + 4z = 13, 3x + 4y + 5z = 40$
6. Solve the system of equations by Gauss-Seidel method
 $4x - y + z = 7, -x + 3y + 2z = 4, x + y + 5z = 6.$
7. Solve the following electrical network equations using Gauss Seidal
 $10x - 5y + 3z = 7, -5x + 15y - 6z = 3, 3x - 6y + 9z = 5$
8. Solve the following electrical network equations using Gauss Seidal
 $5x + 2y + z = 12, x + 4y + 2z = 15, x + 2y + 5z = 0.$
9. A drone delivery system relies on intersecting flight paths. The following three equations describe three flight corridors in 3D space: $x + y + z = 6, 2x + 3y + z = 10, x + 2y + 2z = 9$ Do these three flight paths intersect at a single point (i.e., is the system consistent)? Interpret the geometric meaning.
10. A warehouse tracks three items: A, B, and C. The records show: 2 units of A and 1 unit of B cost Rs.20; 1 unit of A and 2 units of C cost Rs.25; 3 units of B and 4 units of C cost Rs.10. Let x, y, z be the prices of A, B, and C respectively. $2x + y = 20, x + 2z = 25, 3y + 4z = 10$ Is the system consistent? Show whether there is a contradiction.